



and formerly-used defence sites and clearing active, test and training ranges of exposed UXOs.

Another laser system designed for the disposal of UXOs is Laser Avenger system from Boeing Combat Systems (Fig. 14). Boeing successfully tested a one-kilowatt solid-state laser weapon mounted on a converted anti-aircraft vehicle in Redstone Arsenal in Huntsville Alabama, USA, in September 2009, by neutralising multiple types of IEDs including large-calibre artillery munitions, smaller bomblets and mortar rounds (Fig. 15).

The system was operated at safe distances from the targets under a variety of conditions including different angles and ranges. The test followed earlier field tests in which Laser Avenger was used to neutralise five targets representing UXOs and IEDs, and subsequent to that used to shoot down UAVs.

The ordnance neutralisation test was carried out in 2007 with an earlier relatively lower power version of the system. The UAV shoot-down tests were carried out in 2008 and 2009 at White Sands Missile Range, New Mexico, USA, in which Laser Avenger shot down a small unmanned aerial vehicle in each event.

Laser Avenger used a one-kilowatt solid-state laser system mounted on a military Humvee, which is usually equipped with Stinger anti-aircraft missiles. During the tests, Laser Avenger's advanced targeting system acquired and tracked three small UAVs flying against a complex background of mountains and desert, shooting down one of the UAVs.

Rafael's THOR (Fig. 16) is a laser ordnance neutralisation system with an integrated remotely-controlled weapon station. The system comprises a high-energy laser along with its beam director and a coaxial 12.7mm M2 machine gun. THOR uses an air-cooled 700-watt



Fig. 15: Ordnance neutralisation using Laser Avenger



Fig. 16: THOR laser IED neutralisation system



Fig. 17: Laser Ordnance Disposal System

solid-state laser that offers continuous laser engagement with no cool-down time requirement.

The upgraded version of THOR employs a two-kilowatt water-cooled laser with a continuous duty cycle. M2 machine gun is used to perform twin functions, as a standoff disrupter, destroying fusing, thick-cased munitions and booby traps and to provide accurate, direct fire upon enemy forces and targets in either an offensive or defensive role. This dual capability enables THOR to be used for offensive and defensive purposes, as well as for safe stand-off removal of explosive obstacles by laser-directed energy or projectile kinetic energy.

While use of directed high-energy laser gives the operator

the ability to neutralise the IEDs' content by means of burning, deflagrating or detonating the explosive, the machine gun may be used to neutralise the IEDs by targeting the operating device by cutting a wire or detonating cord. The IEDs can be subsequently retrieved by a robot for further neutralisation and investigation.

The system is modular and can be installed on a variety of vehicles and weapon stations, as an add-on system. THOR-like other laser based ordnance disposal systems avoid collateral damage usually associated with other neutralisation procedures.

Laser based systems have highly-minimised associated collateral damage due to explosive burn-out or low-order detonation. In all laser based ordnance disposal systems, directed energy from the laser rapidly clears UXOs and defeats IEDs by inducing a low-order burning or deflagration reaction in the explosive fill at safe stand-off ranges.

Yet another laser based ordnance neutralisation system developed by Defence Research and Development Organisation (DRDO), India, is Laser Ordnance Disposal System (LORDS) (Fig. 17). The system employs a one-kilowatt fibre laser and is mounted along with its support systems on a light security vehicle made by TATA Motors.

The support systems mainly include the thermal management system for waste heat removal, electrical generator to enable standalone operation and command/control system assisted by a CCD camera and a laser range finder for acquisition and beam pointing. The system with its effective range of 30 metres to 250 metres has been successfully field-tested for remote disposal of surface-laid UXOs, mines, directional mines and IEDs.

*To be continued...*